

REST API Design Best Practices

Well-designed APIs are intuitive, consistent, and resilient. This document covers essential patterns for building production-grade REST APIs.

Resource Naming Conventions

Use plural nouns for collections (/users, /orders). Use path parameters for resource identifiers (/users/123). Nest related resources logically (/users/123/orders). Avoid verbs in URLs; HTTP methods convey the action. Use kebab-case for multi-word resources (/user-profiles).

HTTP Status Codes

Use appropriate status codes: 200 for success, 201 for creation, 204 for deletion with no content. 400 for bad requests, 401 for unauthenticated, 403 for unauthorized, 404 for not found, 409 for conflicts, 422 for validation errors. 500 for server errors.

Code	Meaning	Usage
200	OK	Successful GET, PUT, PATCH
201	Created	Successful POST creating a resource
204	No Content	Successful DELETE
400	Bad Request	Malformed request syntax
401	Unauthorized	Missing or invalid authentication
404	Not Found	Resource does not exist
422	Unprocessable	Validation errors (FastAPI default)
429	Too Many Requests	Rate limit exceeded

Pagination and Filtering

Always paginate collection endpoints. Use cursor-based pagination for real-time data (avoids page drift). Use offset/limit for static datasets. Support filtering via query parameters (/users?role=admin&active=true). Return pagination metadata (total_count, next_cursor, has_more).